

1) perception

مسألة باطون 933020

2.

a) extended vectors are: $\theta_t = \begin{bmatrix} w_t \\ b_t \end{bmatrix}$, $\phi_i = \begin{bmatrix} x_i \\ 1 \end{bmatrix}$

margin condition $\rightarrow \langle w_t, x_i \rangle + b_t = \langle \theta_t, \phi_i \rangle$

$\rightarrow$ mistake: $y_i \langle \theta_t, \phi_i \rangle \leq 0$

$\rightarrow$ update rule = $\begin{bmatrix} w_{t+1} \\ b_{t+1} \end{bmatrix} = \begin{bmatrix} w_t \\ b_t \end{bmatrix} + y_i \begin{bmatrix} x_i \\ 1 \end{bmatrix}$

or $\theta_{t+1} = \theta_t + y_i \phi_i$

b)

$$\langle \theta_{t+1}, \theta^* \rangle = \langle \theta_t + y_i \phi_i, \theta^* \rangle = \langle \theta_t, \theta^* \rangle + y_i \langle \phi_i, \theta^* \rangle$$

every sample must satisfy

that: $y_i \langle \phi_i, \theta^* \rangle \geq \gamma$

$$\Rightarrow \langle \theta_{t+1}, \theta^* \rangle \geq \langle \theta_t, \theta^* \rangle + \gamma$$

c)

$$\|\theta_{t+1}\|^2 = \|\theta_t + y_i \phi_i\|^2 = \|\theta_t\|^2 + 2 y_i \langle \theta_t, \phi_i \rangle + \|y_i \phi_i\|^2$$

$$\left. \begin{array}{l} \text{updated so prediction was wrong} \\ \text{so } y_i \langle \theta_t, \phi_i \rangle \leq 0 \\ \|y_i \phi_i\|^2 = \|\phi_i\|^2 = \|x_i\|^2 + 1 \rightarrow \\ \|x_i\| \leq R \end{array} \right\} \Rightarrow \|\theta_{t+1}\|^2 < \|\theta_t\|^2 + 0 + R^2 + 1$$

$$\|x_i\| \leq R$$

d) from $\theta_0 = 0$, if we do T updates

$$(b) \rightarrow \langle \theta_T, \theta^* \rangle \geq T\gamma$$

$$(c) \rightarrow \|\theta_T\|^2 \leq T(R+1)^2$$

$$\text{Cauchy}, \langle \theta_T, \theta^* \rangle \leq \|\theta_T\| \|\theta^*\|$$

$$\Rightarrow T\gamma \leq \|\theta_T\| \|\theta^*\| \leq \sqrt{T(R+1)^2} \|\theta^*\|$$

$$\Rightarrow T^2 \gamma^2 \leq T(R+1)^2 \|\theta^*\|^2$$

$$\Rightarrow T \leq \frac{(R+1)^2 \|\theta^*\|^2}{\gamma^2} \quad \checkmark$$

Requirement 1, is also proven. $\checkmark$

e)

R , $\|\theta^*\|$ and γ are all fixed the T 's bounded

by a finite constant $\rightarrow$ algorithm can't

make infinite updates $\rightarrow$ it must eventually

stop making mistakes $\rightarrow$ converge to a finite number steps

3. what happens? the algorithm will always have at least one misclassified point where $y_i \langle \theta_t, \phi_i \rangle \leq 0 \rightarrow$ weight will ~~not~~ always update infinitely $\rightarrow$ boundary will oscillate for ever

guarantee? No. it will only converge if the data is linearly separable cuz the proof relies on existence of a perfect separator with the positive margin $\gamma > 0$

2)

$$\frac{\partial L}{\partial w_j} = \sum_i \frac{\partial L}{\partial z_i} \cdot \frac{\partial z_i}{\partial w_j}$$

and we know that $z_i = \sum_{k=0}^{K-1} w_k x_{i+k}$

$$\rightarrow \frac{\partial z_i}{\partial w_j} = x_{i+j}$$

$$\rightarrow \frac{\partial L}{\partial w_j} = \sum_i \frac{\partial L}{\partial z_i} x_{i+j} = \sum_i \delta_i x_{i+j}$$

the exact same form in convolution, where we had
 $\text{output}_i = \sum_j \text{filter}_j \cdot \text{input}_{i+j}$ so calculating gradient
of weights is applying a 1D convolution where
input x is convolved with loss gradient $\frac{\partial L}{\partial z}$ (as filter, same).

3) a) effective receptive field size F_{eff} of this detailed
filter is $F_{\text{eff}} = F + (F-1)(D-1) = FD - D + 1$

and the output size without padding ($p=0$) and with

stride ($s=1$) is: $\text{out}_{\text{size}} = \text{in}_{\text{size}} - \text{filter}_{\text{size}} + 1$

$$\rightarrow \text{output Height} = M - F_{\text{eff}} + 1 = M - DF + D$$

$\rightarrow$ Dim of output tensor is

$$(M - DF + D) \times (M - DF + D)$$

b) dilated convo computes dot products over elements by step of D , basically like doing a standard convo using a sparse kernel. we also know that $K \otimes A$ replaces $K_{i,j}$ in K with $K_{i,j}A$

finally to ensure each weight is followed by $D-1$ zeros in both row and columns, A would be:

$$A = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}_{D \times D}$$

c) layer 1: $F_1 = w-1$, dilation $D_1 = d-1$

" 2: $F_2 = w$, $D_2 = d$ / " 3: $F_3 = w+1$, $D_3 = d+1$

from part A, $F_{\text{eff}} = (F-1)D+1 \rightarrow F_{\text{eff},1} = w-2d+3$

$F_{\text{eff},2} = wd-d+1$, $F_{\text{eff},3} = wd+w+1$, and we have

$$R_{\text{total}} = \sum_{l=1}^L (F_{\text{eff},l} - 1) + 1 = 3wd - 3d + 3 = 3(wd - d + 1)$$

$\rightarrow 3(wd - d + 1) \times 3(wd - d + 1)$ pixels

H)

1. a) output is $z = \sigma(w^{(t)} v + b)$, let the pre activation be $a = w^{(t)} v + b$

$$\frac{\partial L}{\partial v_m} = \sum_n \frac{\partial L}{\partial z_n} \frac{\partial z_n}{\partial a_n} \frac{\partial a_n}{\partial v_m} = \sum_n \frac{\partial L}{\partial z_n} (2n \cdot (1 - 2n)) w_{n,m}^{(t)}$$

this was for layer 5, fully connected

for flatten layer, layer 4, there is no operation but a mapping of indices

$$\rightarrow \frac{\partial L}{\partial p_{i,j}^{(3)}} = \frac{\partial L}{\partial \sigma_{2(i-1)+j}} = \sum_n \frac{\partial L}{\partial z_n} [2n \cdot (1 - 2n)] w_{n, 2(i-1)+j}^{(t)}$$

$v_m = p_{i,j}^{(3)}$

b) now, we must back propagate through the average pooling layer and the second convo layer

layer 3: gradient distributes equally

$$\rightarrow \frac{\partial L}{\partial p_{a,y}^{(2)}} = \frac{1}{H} \frac{\partial L}{\partial p_{\{x/2\}, \{y/2\}}^{(3)}}$$

$$\text{layer 2: } \frac{\partial L}{\partial p_{i,j,k}^{(1)}} = \sum_{u=1}^2 \sum_{v=1}^2 \frac{\partial L}{\partial p_{i-u+1, j-v+1}^{(2)}} w_{k,u,v}^{(2)}$$

now we substitute the pooling 6

$$\rightarrow \frac{\partial L}{\partial p_{i,j,k}^{(1)}} = \sum_{u=1}^2 \sum_{v=1}^2 \left(\frac{1}{H} \frac{\partial L}{\partial p_{\left\{\frac{i-u+1}{2}\right\}, \left\{\frac{j-v+1}{2}\right\}}^{(3)}} \right) w_{k,u,v}^{(2)}$$

2.
derivative of loss w.r.t convo filter is just convo
of the layers input with its output gradient

$$\rightarrow P_{x,y,f}^{(1)} = \sum_c x_{a,y,c} w_{f,c,1,1}^{(1)}$$

$$\text{chain rule} \rightarrow \frac{\partial L}{\partial w_{f,K,1,1}^{(1)}} = \sum_{x=1}^2 \sum_{y=1}^2 \underbrace{\frac{\partial L}{\partial P_{a,y,f}^{(1)}}}_{\text{derived in part before}} x_{a,y,K}$$

3.

similarly, we have :

$$\frac{\partial L}{\partial w_{K,(i,j)}^{(2)}} = \sum_{x=1}^2 \sum_{y=1}^2 \frac{\partial L}{\partial P_{a,y}^{(2)}} P_{x+i-1, y+j-1, K}$$

5)

$K(x, y)$ is a valid PSD kernel if it can be written as $K(x, y) = \langle \phi(x), \phi(y) \rangle_H$

where ϕ is map and H is some feature space

also we know the valid kernel are closed under, positive scaling, multiplication, addition and e^x operations.

1. $K(x, y) = \cos(x - y) = \cos x \cos y + \sin x \sin y$

if we define $\phi(x) = [\cos x, \sin x]^T$

$\rightarrow K(x, y) = \phi(x)^T \phi(y)$, in $\mathbb{R}^2 \rightarrow$ PSD $\checkmark$

2.

$K = \cos(x^2 - y^2) = \cos x^2 \cos y^2 + \sin x^2 \sin y^2$

$\phi = [\cos x^2, \sin x^2]^T \rightarrow K = \langle \phi(x), \phi(y) \rangle \rightarrow$ PSD $\checkmark$

3.

$K(x, y) = \frac{1}{x+y} = \int_0^\infty e^{-(x+y)t} dt = \int_0^\infty e^{-xt} e^{-yt} dt$

$\phi(x)(t) = e^{-xt} \rightarrow K = \langle \phi(x), \phi(y) \rangle_{L_2}$

where L_2 is $[0, \infty)$

4. $K = \cos \angle(x, x') = \frac{x^T x'}{\|x\| \|x'\|}$

$$\phi(x) = \frac{x}{\|x\|} \rightarrow K(x, x') = \phi(x)^T \phi(x') \rightarrow \text{PSD} \checkmark$$

5.

$$K = \exp(-\lambda (\sin(x' - x))^2)$$

we know that $\sin^2(x' - x) = \frac{1 - \cos 2(x' - x)}{2}$

$$= \frac{1}{2} - \frac{1}{2} \left[\cos 2x \cos 2x' + \sin 2x \sin 2x' \right]$$

if we consider $\phi(x) = [\cos 2x, \sin 2x]^T$

$$\rightarrow \sin^2(x' - x) = \frac{1}{4} \|\phi(x) - \phi(x')\|^2$$

$$\rightarrow K = \exp\left(-\frac{1}{4} \lambda \|\phi(x) - \phi(x')\|^2\right)$$

exactly the standard Gaussian RBF kernel. PSD ✓

6)

to find dimension of $\phi(x)$ for $K(x, x') = (x \cdot x' + c)^d$

we count the linearly independent terms in its expansion.

a general monomial: $x_1^{q_1} \dots x_n^{q_n}$, $q_i \geq 0$, $\sum_{j=1}^n q_j \leq d$

so we can say for some $q_{N+1} \geq 0 \rightarrow \sum_{j=1}^N q_j + q_{N+1} = d$

so we are distributing d identical items among $n+1$

distinct bins $\rightarrow \text{Dimension} = \binom{d+(n+1)-1}{(n+1)-1} = \binom{n+d}{n} = \binom{n+d}{d}$

if we use standard Binomial theorem:

$$K(x, x') = (x \cdot x' + c)^d = \sum_{i=0}^d \binom{d}{i} (x \cdot x')^i c^{d-i}$$

Substitute $K_i(x, x')$ into $K(x, x')$

$$K = \sum_{i=0}^d \binom{d}{i} c^{d-i} K_i(x, x')$$

$$\implies \text{assigned weight is } w_i = \binom{d}{i} c^{d-i}$$

So if $c > 0$:

add $w_i > 0$ so PSD property remains

$c < 1$:

the c^{d-i} term grows as i decreases so kernel places higher relative weight on lower degree features.

and also we can say for $0 < c < 1$, c acts as a hyperparameter that acts as control trade off between high and low order features.

7)

1. Bayes classifier $h^*(x)$ outputs the most probable class label for a given input x to minimize probability of misclassification:

$$h^*(x) = \arg \max_{y \in \{-1, +1\}} P(y|x)$$

$\implies$ dimension is 1

Kernel is PD if for any finite set $x_1, \dots, x_N$
 and any $c_1, \dots, c_N \in \mathbb{R}$, $\sum_{i=1}^N \sum_{j=1}^N c_i c_j K^*(x_i, x_j) \geq 0$
 $\rightarrow \sum_{i=1}^N \sum_{j=1}^N c_i c_j h^*(x_i) h^*(x_j) = \left(\sum_{i=1}^N c_i h^*(x_i) \right)^2 \geq 0$

which always holds $\rightarrow$ PD $\checkmark$

2. input x_i is mapped to 1D feature space

$z_i = \phi(x_i) = h^*(x_i) \in \{-1, 1\}$, SVM looks for 1D

weight w , bias b that: $y_i (w z_i + b) \geq 1$

the S mapped to feature space is linearly separable
 if and only if there is zero classification error
 with Bayes $\rightarrow y_i = h^*(x_i)$ for all i
 assuming this holds, SVM:

$y_i (w y_i + b) \geq 1 \rightarrow w + y_i b \geq 1$

so we must have $w + b \geq 1$, $w - b \geq 1$ while
 $\frac{1}{2} \|w\|^2$ is minimized $\rightarrow w = 1, b = 0$

$\rightarrow f(x) = \text{sgn}(1 \cdot h^*(x) + 0) = h^*(x) \checkmark$

$\gamma = \frac{1}{\|w\|} = 1 \checkmark$

all m points are support $\checkmark$
 all lie on boundary

cuz SVM

f is h^* itself

generalization
 error = Bayes
 error rate

3.

suppos $h: X \rightarrow \mathbb{R}$ is some function and $K = h(a_i)h(a_j)$
there is no special condition on h , as long as it's
real valued. Proof:

symmetry:

$$K(a_i, a_j) = h(a_i)h(a_j) = h(a_j)h(a_i) = K(a_j, a_i)$$

pf:

as established in part 1, Gram sum simplifies

to $\left(\sum_i c_i h(a_i) \right)^2 \geq 0$, h only produces real
numbers and any linear combination is real hence
the square remains ≥ 0 .

8.)

1. in hard margin svm:

$$\min_{w, b} \frac{1}{2} \|w\|^2 \text{ s.t. } y_i (\langle w, a_i \rangle + b) \geq 1 \quad \forall i$$

for such problem to have solution, the feasible set
must be non-empty $\rightarrow$ at least one (w, b) pair
must satisfy the condition for all training points
and the $y_i (\langle w, a_i \rangle + b) \geq 1$ is definition of linearly separable

margin between Boundary and closet datapoint:

$\gamma = \frac{1}{\|w\|}$, and also minimizing $\|w\|$ is the same as minimizing any monotonic function of it like $\frac{1}{2}\|w\|^2$, and by that we see that objective

perfectly maps to finding the max-margin hyperplane dual:

the Lagrangian is:

$$L(w, b, \alpha) = \frac{1}{2}\|w\|^2 - \sum_{i=1}^m \alpha_i \left[y_i (\langle w, x_i \rangle + b) - 1 \right]$$

$$\nabla_w L = w - \sum_{i=1}^m \alpha_i y_i x_i = 0 \rightarrow w = \sum_{i=1}^m \alpha_i y_i x_i$$

$$\frac{\partial L}{\partial b} = 0 \rightarrow \sum_{i=1}^m \alpha_i y_i = 0$$

substitute
back

$$\max_{\alpha} \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$$

$$\text{s.t. } \alpha_i \geq 0 \quad \forall i, \quad \sum_{i=1}^m \alpha_i y_i = 0 \quad \checkmark$$

using the dual solution α , the normal vector is

$w = \sum \alpha_i y_i x_i$, for a new point x_i :

$$f(x) = \text{sgn}(\langle w, x \rangle + b) = \text{sgn} \left(\sum_{i=1}^m \alpha_i y_i \langle x_i, x \rangle + b \right)$$

$$\text{where } b = y_K - \sum \alpha_i y_i \langle x_i, x_K \rangle$$

2.

if data is not strictly separable, we introduce slack variable ε_j :

$$\min_{w, b, \varepsilon} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^m \varepsilon_i$$

the ε allows points to cross margin boundary ($0 < \varepsilon < 1$) or decision ($\varepsilon > 1$), they measure the degree of violation

and C is the penalty hyperparameter, to control the trade-off between ~~max~~ margin and min classification error.

if we define:

$$L = \frac{1}{2} \|w\|^2 + C \left[\sum_{i=1}^m \varepsilon_i - \sum \alpha_i (y_i (\langle w, a_i \rangle + b) - 1 + \varepsilon_i) \right]$$

$$- \sum_{i=1}^m \alpha_i \varepsilon_i \Rightarrow \nabla_w L = 0 \rightarrow \sum \alpha_j y_i a_i = 0$$

$$\frac{\partial L}{\partial b} = \sum \alpha_i y_i = 0, \quad \frac{\partial L}{\partial \varepsilon_j} = 0 \rightarrow C - \alpha_i = \alpha_i$$

substitute

$$\max_{\alpha} \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \langle a_i, a_j \rangle$$

constraints were: $\xi_j > 0$, $\xi_j > 1 - y_i (\langle w, a_i \rangle + b)$

combine, $\xi_j > \max \{0, 1 - y_i (\langle w, a_i \rangle + b)\}$

we are minimizing the objective function:

$\frac{1}{2} \|w\|^2 + C \sum \xi_j$ so the optimal ξ will be as small

as possible $\rightarrow \xi_j = \max \{0, 1 - y_i (\langle w, a_i \rangle + b)\}$

(exactly the hinge loss by the way)

substitute
 ξ_j and
derive $\min_{w, b} \underbrace{\frac{1}{2mC}}_{\xi} \|w\|^2 + \frac{1}{m} \sum_{i=1}^m \max \{0, 1 - y_i (\langle w, a_i \rangle + b)\}$ ✓